



# **Progress in Computational Modelling for Concrete Durability and Its Integration with Artificial Intelligence and Life-Cycle Assessment**

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# Screenshot of Headline

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REVIEW



## Progress in Computational Modelling for Concrete Durability and Its Integration with Artificial Intelligence and Life-Cycle Assessment

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# MAIN FINDINGS AND ARGUMENTS

- Traditional durability design methods are overly prescriptive and often fail to reflect real long-term concrete behavior.
- Computational durability models provide more accurate predictions of carbonation, chloride ingress, freeze-thaw damage, and chemical attack.
- Artificial intelligence improves prediction accuracy by identifying patterns in large experimental and field datasets.
- Integrating AI with durability models enables performance-based design instead of rule-based design.
- Coupling durability modelling with Life-Cycle Assessment allows more realistic estimation of service life and embodied carbon.

# BROADER IMPLICATIONS FOR SOCIETY

- AI-driven durability prediction reduces premature infrastructure failure.
- More accurate service-life estimation leads to lower long-term environmental impact.
- Improved durability reduces maintenance frequency and public repair costs.
- Integration of AI and LCA supports sustainable development and climate goals.
- Infrastructure becomes more resilient to environmental and climate-related stressors.

# WHY IT MATTER IN CONSTRUCTION?

- AI bridges the gap between lab tests and real-world performance.
- Supports smarter material selection and mix design.
- Enhances asset management and long-term infrastructure planning.
- Aligns construction practices with sustainability and carbon-reduction goals.